

Let $\mathfrak{g}$ be a Lie algebra. Let

$$C^k = \wedge^k \mathfrak{g}^*, \quad \delta: C^k \rightarrow C^{k+1},$$

$$\delta c \mapsto (X_0, \dots, X_n) \mapsto \sum_{i < j} (-1)^{i+j} c([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_n).$$

Let

$$H^k(\mathfrak{g}; \mathbb{R}) = H^k(C^k, \delta).$$

By averaging, this agrees with the de Rham cohomology of the associated compact connected Lie group.

Note:

if $c \in \mathfrak{g}^*$ then $\delta c(X_0, X_1) = -c([X_0, X_1])$, so

$$H^1(\mathfrak{g}; \mathbb{R}) = \text{Ann}[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}^*;$$

in particular, H^1 of a semisimple Lie algebra is zero.

Thm (Whitehead): If $\mathfrak{g}$ is semisimple then $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$.

Prop: If $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$ then every symplectic action is Hamiltonian.

Proof: Let $\Psi: G \rightarrow \text{Symp}(M, \omega)$ be a symplectic action. Since $H^1 = 0$, $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$. We need a moment map

$$\begin{array}{ccc} \mathfrak{g} = [\mathfrak{g}, \mathfrak{g}] & \xrightarrow{\partial\Psi} & \mathcal{X}^{\text{Ham}} \\ & \searrow & \nearrow \\ & C^\infty(M) & \end{array}$$

since commutators of symplectic vector fields are Hamiltonian

For each basis vector $X \in \mathfrak{g}$, choose a lift $\tau(X) \in C^\infty(M)$ with $\partial\Psi(X) = X_{\tau(X)}$.

This may not be a Lie algebra homomorphism. We have

$$\begin{aligned} \partial(\tau([X, Y])) &= \omega([X, Y]^\#, \cdot) \\ &= \omega(-[X^\#, Y^\#], \cdot) \\ &= \partial\{\tau(X), \tau(Y)\}. \end{aligned}$$

Hence,

$$\tau([X, Y]) - \{\tau(X), \tau(Y)\} = c(X, Y) \in \mathbb{R}$$

is constant. Clearly, $c \in \wedge^2 \mathfrak{g}^*$. We have

$$\begin{aligned} \delta c(X, Y, Z) &= -c([Y, Z], X) + c([X, Z], Y) - c([X, Y], Z) \\ &= c(X, [Y, Z]) + c(Y, [Z, X]) + c(Z, [X, Y]) \\ &= \tau([X[YZ]] + [Y[ZX]] + [Z[XY]]) \\ &\quad - \{\tau(X), \tau([Y, Z])\} - \{\tau(Y), \tau([Z, X])\} - \{\tau(Z), \tau([X, Y])\} \\ &= 0, \end{aligned}$$

$\uparrow$
 $= \{\tau(Z), \tau(X)\}$
up to a constant $c(X, Y)$

by the Jacobi identity for $[\cdot, \cdot]$ and $\{\cdot, \cdot\}$.

Hence, $c \in H^2 = 0$ and $c = \delta b$ for some $b \in \mathfrak{g}^*$. So

$$c(X, Y) = -b([X, Y]).$$

Let $\mu^*: X \mapsto \tau(X) + b(X)$. Then μ^* is a Lie algebra homomorphism:

$$\begin{aligned} \mu^*([X, Y]) &= \tau([X, Y]) + b([X, Y]) \\ &= \underbrace{c(X, Y)}_{-b([X, Y])} + \{\tau(X), \tau(Y)\} + b([X, Y]) = \{\tau(X), \tau(Y)\}. \quad \square \end{aligned}$$